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Machine learning assisted innovative strategy for experimental validation of functional properties of lead-free ceramics

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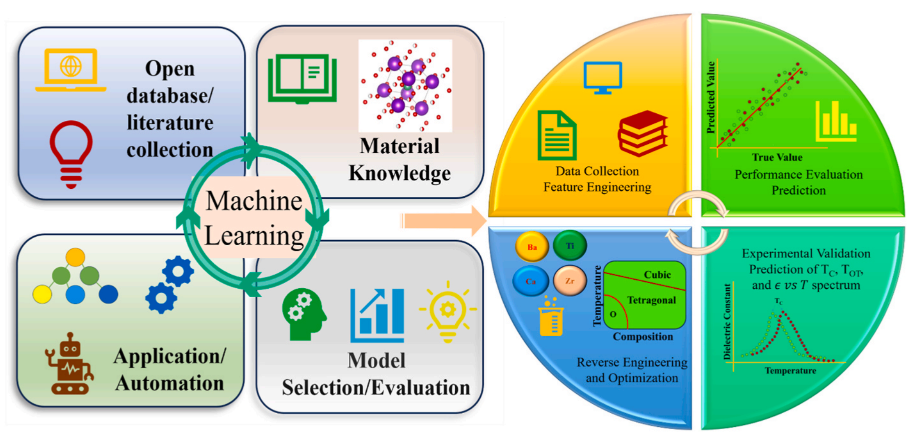
HIGHLIGHTS

- Implemented machine learning (ML) to accelerate the search for lead-free ceramics.
- Predicted T_C, T_{OT}, and dielectric temperature spectrum for lead-free ceramics using machine learning.
- Explored the ability of ML models in predictive tasks and synthesis process optimization for lead-free compositions.
- Experimental validation was performed for the developed ML models with prepared Li-doped $BaTiO_3$ compositions.

GRAPHICAL ABSTRACT



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ABSTRACT

Lead-free ferroelectric ceramics are particularly prominent in the energy sector because of their interesting multi-domain properties. In the present study, three supervised machine learning models (random forest, XG-boost, and SVR-rbf) were implemented to predict and optimize the functional properties of lead-free ceramics. The machine learning models are successfully able to predict the T_C, T_{OT}, and dielectric temperature spectrum of the newly considered multi-component ferroelectric $(Ba_{1\text{-}x\text{-}y}Ca_xLi_y)$ $(Ti_{1\text{-}u\text{-}v\text{-}w}Zr_uSn_vHf_w)$ O_3 system with a certain tolerance ($\Delta T_C = 5.87$ °C, and $\Delta T_{OT} = 7.53$ °C) to achieve the accelerated prediction. Additionally, engineered features are utilized as inputs, derived from material science insights for the supervised machine learning models, and essential features among them are identified. For experimental validation, Li-doped $BaTiO_3$ samples were prepared via a solvothermal route, and properties were predicted with a maximum error of 11.4 % in T_C for the as-prepared samples. Finally, the reverse engineering analysis of the lead-free ceramics was performed with the help of elemental descriptors to optimize the synthesis process by tuning the stoichiometry of the composition.

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1. Introduction

The ferroelectric ceramics exhibit outstanding piezoelectric and dielectric characteristics; hence, they are considered as promising candidates in the sphere of energy harvesting, sensors, and smart materials [1,2]. It is practicable to achieve ultrahigh piezoelectric constant in lead-free ceramics via insightful compositional design and experimental optimization, as demonstrated in several studies [3–6]. While lead-based materials performed better in piezoelectric applications, their use was limited because of environmental concerns. Among lead-free ferroelectric materials, $BaTiO_3$ - based (BT) ceramics were extensively studied to understand their crystal structure, phase transitions, and corresponding physical mechanisms for high piezoelectric properties [2,7–11]. Recently, it has been shown that upon introducing specific cations to substitute for A-site (Ca^{2+}) and B-site (Zr^{4+}) in $BaTiO_3$, a high value of d_{33} (>520 pC/N) was achieved for $Ba_{0.85}Ca_{0.15}Zr_{0.1}Ti_{0.9}O_3$ (BCZT) [3] ceramics and Tm-modified BCZT [12]. As lead-free ferroelectric materials are increasingly prevalent because of their high piezoelectric characteristics and biocompatibility [13–15], many researchers are exploring chemically modified lead-free multiferroic materials for their diverse applications in the industry [2,16]. The barium titanate–based ceramics are significant in energy harvesting because they possess lower dielectric loss, high dielectric constant, and excellent piezoelectric performance.

The discovery of new perovskite materials with enhanced ferroelectric properties has historically been directed by intuition and trial methods. In today's era of machine learning (ML), it is achievable to explore the vast space of chemical compositions to identify potential candidates for smart-materials applications [4,17,18]. Machine learning has already been implemented in several sectors to build informative models and solutions for many real-time problems [19–26]. ML models are practically data-driven and can potentially extract the hidden pattern from the given input data to elucidate the underlying mechanism [4]. In the newly introduced stream of materials informatics, machine learning algorithms were most useful in investigating the composition-property relationships [27–31]. Within the energy domain, several researchers have employed machine learning models to facilitate the prediction of functional and ferroelectric properties exhibited by lead-free ceramics [5,19,20]. Additionally, a growing amount of literature exists analyzing critical descriptors essential for predicting the ferroelectric properties of perovskite materials [19,32]. Artificial intelligence algorithms allow researchers to make informative choices based on insights from previous experimental data and computational studies [33]. With the implementation of diverse ways of feature engineering, the property prediction capabilities of the ML models increased magnificently, but the black-box effect in machine learning algorithms is usually criticized for not allowing researchers to achieve physical insights into the possible dynamics. To tackle the issue of the black-box effect in machine learning, researchers invented interpretable machine learning strategies such as the sure independence screening and sparsifying operator (SISSO) method to extract the essential features and their relation with the target properties [34]. Although machine learning algorithms can successfully solve many real-time problems in science, machine learning applications in materials science are still in the beginning stages, and ample opportunities exist for more insightful research [5]. Machine learning models combined with experimental studies already proved beneficial for research as they substantially helped to shorten the experimentation time, reduction in cost, and optimization of the processes [4,28,32,35].

It is possible to create predictive ML models for lead-free ceramics with the data available in the form of literature and, by doing so, predict the properties of the compounds with different stoichiometry [35]. The essential task is to collect the data carefully from reliable resources and preprocess it to avoid any potential outliers in the data. Recently, crucial studies have been conducted to accelerate the prediction of $BaTiO_3$--based ceramics for energy applications. He et al. predicted the multi-component phase diagrams and dielectric temperature spectrum for the system $(Ba_{1\text{-}x\text{-}y}Ca_xSr_y)$ $(Ti_{1\text{-}u\text{-}v\text{-}w}Zr_uSn_vHf_w)O_3$ with fine boundary information with the help of regression and classification models [35,36]. These studies were performed with the help of neural network models having Ca^{2+} and Sr^{2+} substituting for the Ba^{2+} site. In the present study, several features were computed for a given compound from elemental property values such as electronegativity, ionic radii, polarizabilities, atomic weight and atomic number of elements, and atomic volume of elements, which are known to be critical in the prediction of ferroelectric properties. Again, very little literature has reported the predictive analysis of solvothermally synthesized piezoelectric perovskites. Therefore, A-site Li-doped $BaTiO_3$ ceramics were synthesized solvothermally for experimental validation of the developed machine-learning models. In addition, the reverse engineering concept was implemented to accelerate the identification of available compounds for a specific range of T_C or T_{OT}, based on the application.

The article demonstrates ML algorithms, which can support monitoring the functional properties. For the same, the three supervised machine learning models were trained (SVR-rbf, XG-Boost, and random forest) to predict the phase transition temperatures (T_C and T_{OT}) and dielectric temperature spectrum of considered $(Ba_{1\text{-}x\text{-}y}Ca_xLi_y)$ $(Ti_{1\text{-}u\text{-}v\text{-}w}Zr_uSn_vHf_w)$ O_3 - based ceramics with low standard deviations (ΔT_C = 5.87 °C, and ΔT_{OT} = 7.53 °C). The proposed model captured the T_{OT} phase transition point in the predicted dielectric temperature spectrum. The SVR-rbf model demonstrated the best results, $R^2 = 0.96$ for T_C, and $R^2 = 0.9$ for T_{OT}. The XG-Boost model is implemented to predict the dielectric temperature spectrum of 17 different compositions with a mean absolute error of 8.1 °C, giving predictions close to the actual value as compared to other models, with very few entries having a high deviation value. The machine-learning models were validated by predicting T_C for Li-doped $BaTiO_3$, with a maximum error of 11.4 %. The essential features and their order of importance were extracted from ensemble-based models. Additionally, a limited amount of available data from the literature restricts the machine learning model's performance; hence bootstrapping technique is implemented to improve the accuracy of the model. Also, the proposed reverse engineering helps identify the available compositions with the value of T_C or T_{OT} lying in a specific range.

2. Machine learning methods

The ML models for the given study are implemented using the "scikit-learn" library based on the python language. Three models were implemented in the study, namely, "SVR-rbf," "XG-Boost," and "random forest," to perform the forecast of the properties. In the present analysis, two different datasets were prepared with a total of 13 features for T_C/T_{OT} prediction and 14 features for the prediction of the dielectric temperature spectrum from the literature. The analysis will classify the two datasets as "Dataset 1" and "Dataset 2" for further discussion. In dataset 1, a total of 114 different lead-free compositions from the literature with their corresponding T_C and T_{OT} values were added. The research task here did not consider compositions having T_{OT} phase transition temperatures less than 0 °*C*. Meanwhile, Dataset 2 contains different compositions (approximately 154 compositions) of lead-free ceramics, with their dielectric temperature spectrum information constituting a total number of entries of 11625. The first few entries of all the datasets are provided in the Supplementary Information (Tables S1–S3). In Dataset 2, the entries for each composition are collected from the experimental results demonstrated in the literature, having an even temperature spacing of 2°*C*, and the remaining features were calculated similarly to Dataset 1. In Dataset 2, the current analysis did not consider the exact temperature range for various compositions, as temperature ranges differ with different literature. The complete list of feature sets is provided in Table 1 for reference, and the features for various compositions were calculated with the help of Equation (1). Within equation (1), X denotes the characteristic property of the relevant perovskite material.

Table 1
Features utilized in the study with their description.

No.	Feature Name	Description and Reference
1	PA	Weighed polarization of elements at A site [37]
2	PB	Weighed polarization of elements at B site
3	ENA	Weighed MB Electronegativity of elements at A site [38]
4	ENB	Weighed MB Electronegativity of elements at B site
5	RA	Weighed Ionic radii of elements at A site [39]
6	RB	Weighed Ionic radii of elements at B site
7	VA	Weighed Atomic volume of elements at A site [40]
8	VB	Weighed Atomic volume of elements at B site
9	At. Comp.	Weighed Atomic mass of composition
10	At. Num.	Weighed Atomic number of compositions
11	CRSA	Weighed Core radii sum of A site elements [38]
12	CRSB	Weighed Core radii sum of B site elements
13	NTC	Effect of doping on T_C ϵ (−1,0,1)
14	T	Temperature (for dielectric spectrum prediction)

The variable x signifies the precise stoichiometric proportion of other ceramic components present within the solid solution.

$$X_{solid\ solution} = \left(\left\{x \times X_{ABO_3}\right\} + \left\{(1-x) \times X_{BT}\right\}\right) \tag{1}$$

Further, to identify the correlation between the different independent features for all the datasets (with different target properties), the Pearson correlation heat-map attribute is implemented for quantification purposes (Fig. 1). The limited availability of dataset entries from the reliable literature motivated us to utilize the bootstrapping technique to find the mean value for prediction and corresponding standard deviation for each composition (Supplementary Information) [41]. Further, the "Root Mean Square Error" (RMSE) score is used for the test dataset to estimate the performance of the implemented machine learning models. Next, in order to tune the models with an appropriate set of hyperparameters, the "GridSearchCV" routine available in the scikit-learn library is utilized with a sufficient sample space of hyperparameters. In "GridSearchCV," the RMSE score is implemented to identify the unique set of hyperparameters for the model's optimal performance. Hyperparameter tuning in machine learning models allows researchers to improve the model's reliability during the prediction of unseen data in diverse applications. The features created in training and test datasets were standardized so that the resulting mean and standard deviation were 0 and 1, respectively, to avoid the magnitude bias among the features. Specific details about the implemented machine learning models can be found in the supplementary information file.

In order to create a train and test dataset, the available dataset is divided in the ratio of 70:30 for further modeling. The training dataset was employed for both model training and hyperparameter tuning purposes to achieve optimal performance. The test dataset is used to

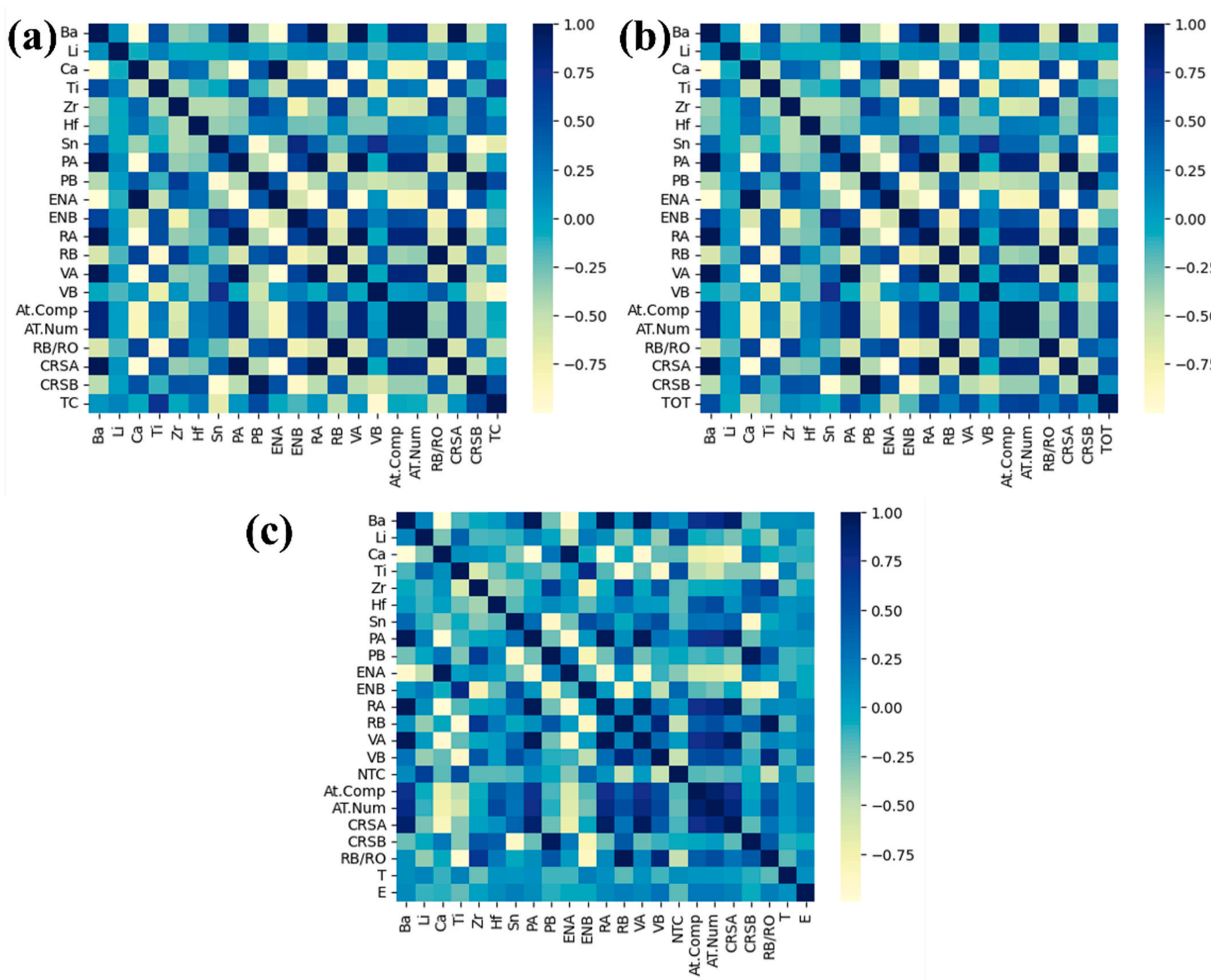

Fig. 1. Correlation matrix of features for (a) T_C, (b) T_{OT} (dataset 1), and (c)ϵ_r.

measure the performance of the final optimized model. The process mentioned above is adopted for Dataset 1 only; as for Dataset 2, a separate test dataset is created to evaluate the prediction capabilities of the model. The features in all the datasets are chosen in such a way that they are able to reflect structural, physical, and chemical properties of the composition in a better way so that the model could be able to develop the relation between the target property and the included independent features. It is important to note that the functional properties of piezoelectric ceramics primarily depend upon compositional factors (elemental stoichiometry variation), structural factors (crystal structure, lattice distortion), electrochemical factors (electronegativities), and bond factors (polarizability). Incorporating these factors in the form of features with the help of feature engineering is essential and holds the potential for possible physical insights. Separate features were calculated for A-site and B-site elements from the stoichiometric formula of the compound and with the help of literature. Among the calculated features, all those were selected for training the machine learning model, which holds a higher probability of affecting the target functional property (Table 1). Ensemble machine learning models were used to identify critical features because of their unique functionality and rank the input features according to their importance.

3. Feature selection and model performance

Pre-processing the information collected from literature and feature engineering is primely important for machine learning modeling in material science. In order to create meaningful feature sets for predictions, feature engineering requires material science insights, which will enable machine learning models to perform much better with higher accuracy. Also, it will allow researchers to find a significant relation between the features and the target properties. In the present study, the feature set considered in Table 1 was created with the initial guesswork from the theory of piezoelectricity and insights about ferroelectricity from published literature [19,20,41]. The current analysis observed that the order of importance for the descriptors varies depending on the target properties. Also, using a limited number of features in the model is essential to reduce the complexity and enhance the performance. To reduce the number of features, the features with low importance scores were eliminated. It is important to generalize the model for predicting the properties of unseen compositions based on $BaTiO_3$; hence each known composition is uniquely described with the help of independent structural, electrochemical, and compositional features. An analysis of the previously identified factors, including compositional, electrochemical, and size-related elements, alongside other potential features, has demonstrated their substantial impact on the designated functional property since these factors are traditionally well-known indicators of ferroelectricity and structural stability in perovskite materials. The features RA and RB were calculated using the ionic radii of the elements after consideration of their co-ordination number in the crystal structure. Also, separate columns were added for elements to capture the stoichiometry of each composition in the feature set.

The training data is utilized to tune the hyperparameters and train the model for the prediction of the test dataset (for T_C and T_{OT}) (Fig. 2). The performance of all three models was measured with RMSE and R^2 metrics. The SVR-rbf model showed optimal performance in the prediction of T_C and T_{OT} concerning R^2 value (Fig. 2). Analysis of the data revealed that the ensemble learning approaches, specifically random forest, and extreme gradient boosting, also demonstrated good performance since their RMSE value for the test dataset is comparatively low (Fig. 3). From Fig. 2, it can be seen that the R^2 value of SVR-rbf is outperforming the R^2 of ensemble models. As already observed in Fig. 2, the positions of the predicted values from the test dataset close to the reference line (y = x line) signify the better performance of the SVR-rbf model. Upon further investigation, the analysis identified that the XG-boost model is more efficient in predicting T_C and T_{OT} for a series doping and dielectric temperature spectrum. As previously reported in a few works of literature, the feature importance obtained in analysis for T_C prediction matches with the explained physics [35,42]. The

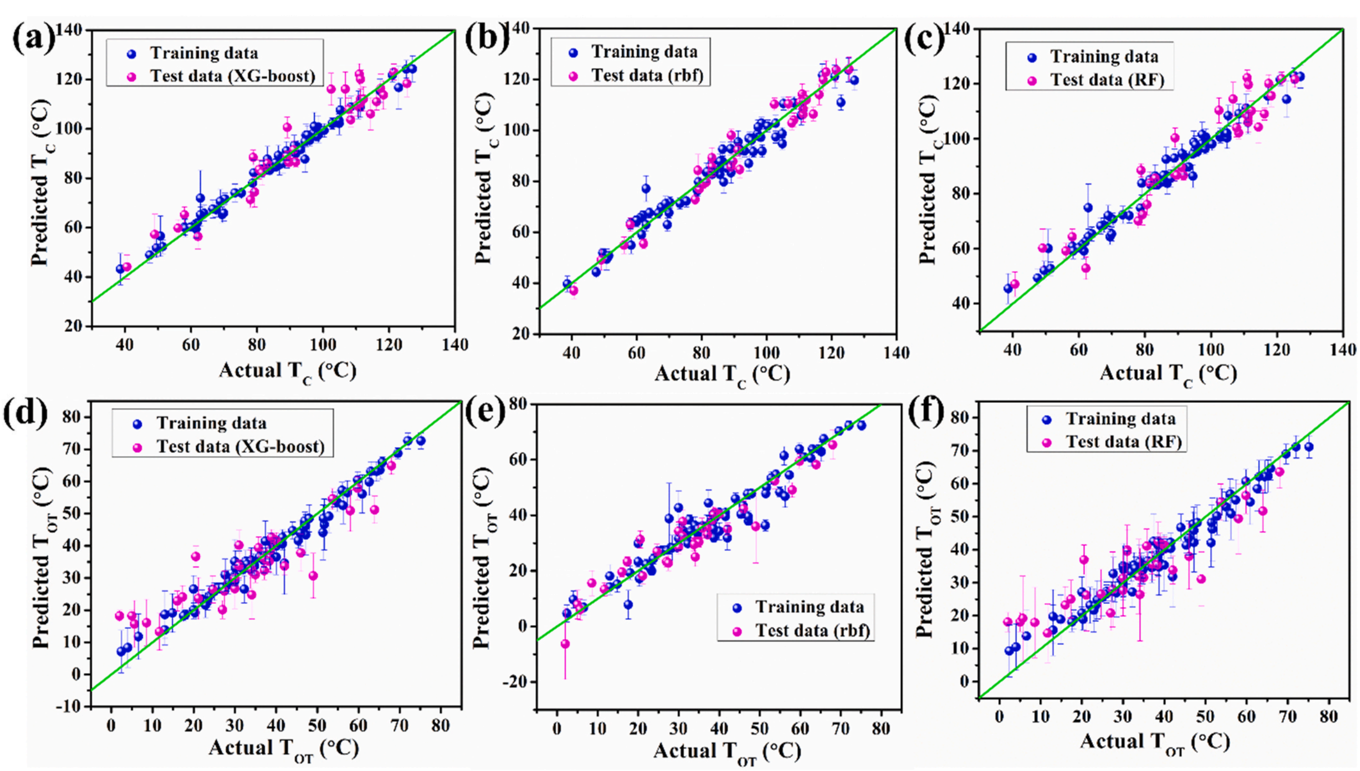

Fig. 2. Prediction performance of ML-models for T_C: (a) XG-Boost, $R^2 = 0.93$, (b) SVR-rbf, $R^2 = 0.96$, (c) RF, $R^2 = 0.92$, and for T_{OT}: (d) XG-Boost, $R^2 = 0.78$, (e) SVR-rbf, $R^2 = 0.90$, (f) RF, $R^2 = 0.76$.

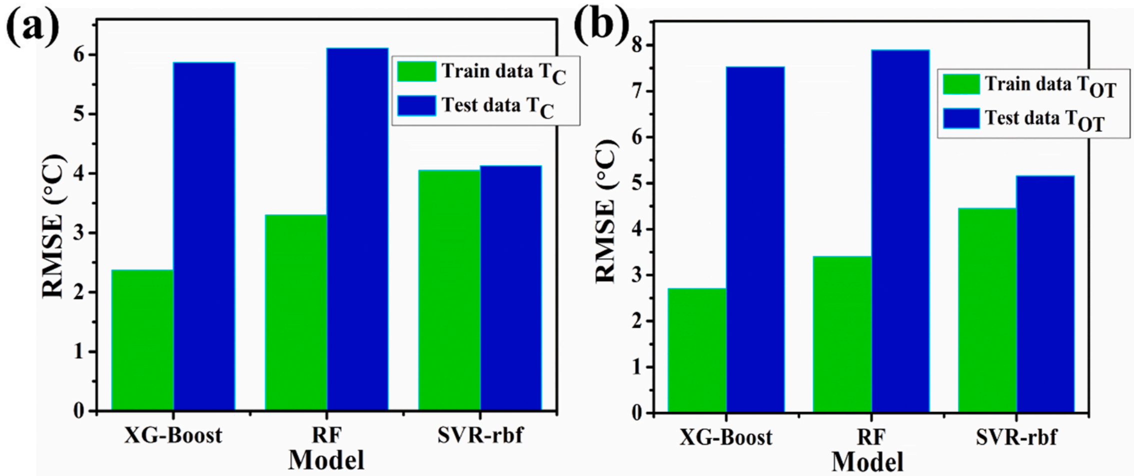

Fig. 3. Performance analysis of various machine learning models for the prediction of (a) T_C and (b) T_{OT}.

investigation into predicting transition temperature (T_C) with ensemble models has identified critical features influencing the outcome. These descriptors primarily include VB, atomic composition, and electrochemical properties of B-site elements [19,32]. Extracting the physical and material science insights from the machine learning model necessitates a meticulous analysis of the feature importance score associated with various target properties. J He et al. [35] also obtained MB electronegativity and core electron radii as critical features in the prediction of T_{OT}. Also, it is important to note that, during the prediction of T_{OT}, A-site features showed significant contributions (CRSA, RA). The contribution of A-site and B-site elements is also visible in the behaviour of T_{OT}. For example, the addition of Ca^{2+} at the A-site is performed to lower the value of T_{OT}, and the addition of other elements, such as Zr^{4+}, Hf^{4+}, and Sn^{4+}, is done to increase the value of T_{OT} [43]. It is already known that the dielectric properties of $BaTiO_3$-ceramics are very sensitive to phase boundary types and the type of doping elements [43]. When compared, doping of Hf^{4+} and Zr^{4+} increases T_{OT} faster than Sn^{4+} for $BaTiO_3$-ceramics. Meanwhile, Sn^{4+} decreased T_C much faster than Zr^{4+} and Hf^{4+} [43]. For $BaTiO_3$-based piezoelectric ceramics, the spontaneous polarization is a result of Ti-atom off-centering within the oxygen octahedron; hence, the features representing the B-site elements will directly influence the piezoelectric and ferroelectric properties [10].

Upon investigation of the results, features representing the atomic volume, atomic weight, atomic number, atomic polarization, and MB electronegativity of the constituent elements in the composition are highly influencing the targeted functional property. Within the category of size factors influencing ferroelectric properties, ionic radii and atomic volume of B-site elements emerged as particularly significant contributors. The heightened importance stems from the crucial role played by B-site elements in the overall behaviour of perovskite compounds. In selecting electrochemical properties, MB electronegativity is preferred over Pauling electronegativity because it depends on ionization potentials and provides different electronegativity values for ions of the same element with different valences [44]. It is observed that average electronegativity does reflect the structural distortion, rotation, and tilt of BO_6 octahedral in perovskite compounds [19]. Factors such as the compound's atomic weight and the constituent elements' polarizability are significant indicators for predicting the ferroelectric properties. The importance of the feature is found out with the help of ensemble models (XG-boost and RF). The results are demonstrated in Figure S1 and Figure S2. During transition temperature (T_C) prediction, features encompassing size factors and the electrochemical properties of B-site elements have been identified as playing a particularly crucial role [45].

The observation from machine learning matches with the experimental observations because elemental doping at B-site other than titanium (Zr^{4+}, Hf^{4+}, and Sn^{4+}) causes a reduction in the Curie temperature. Similarly, Sr^{2+} also causes a decrease in Curie temperature of $BaTiO_3$-based ceramics, but Sr^{2+} doping is not considered in our study [43]. In the case of T_{OT} prediction, the first few important features (from Fig. S1) are ENB, PB, At. Num., At. Comp., and A-site dependent properties, which are also completely in alignment with the insights from the experiments.

To further validate the machine learning models, the present research has attempted to reproduce the composition vs temperature diagrams and dielectric temperature spectrum of the lead-free compositions from the relevant literature (with different stoichiometry not present in the training dataset) and found that they are in good agreement with the actual results as shown in Fig. 4 and Figure (S4, S5), respectively. The XG-Boost model successfully predicted the dielectric temperature spectrum for all 14 series-based compositions, out of which only three compositions have prediction errors above 18 % in T_C value. In order to predict the dielectric temperature spectra and physically explain the factors that affect the shape, peak, and position of dielectric peaks, the current study has incorporated domain knowledge in the feature pool by introducing physical features related to phase transition and polarization. As we have already gained crucial insights about the features that influence T_C and T_{OT}, we can implement them directly to predict the dielectric temperature spectrum, adding one extra independent variable of "temperature" to the dataset. Predicting the temperature spectrum is crucial since predicting curves has numerous applications in materials science, and it is essential to understand the properties of the different materials. Using machine learning, researchers already predicted X-ray absorption spectra [46] and ferroelectric hysteresis loops [47] for material characterization. J He et al. [36] predicted the dielectric temperature spectrum for lead-free ceramics with deep learning models and identified the critical independent features of the spectrum. Also, as it is essential to maintain the continuity between consecutive points in the curve prediction task, it is difficult to achieve in a point-by-point prediction model. In our study, we showed that with the help of ensemble models like XG-boost, the dielectric temperature spectrum for $BaTiO_3$-based ceramics can be predicted. XG-boost model is chosen over SVR-rbf because of the low value of RMSE for the test dataset in the temperature prediction task, as well as the advantage of better reverse engineering predictions. In the present model, we captured T_{OT} as well in the predicted dielectric temperature spectrum, with feature analysis. The features that are used in the dataset influence the dielectric peak positions and the shape of the spectrum. The importance of the feature in predicting the dielectric temperature spectrum is provided in Fig. S2.

Two systems from the literature were considered to validate our

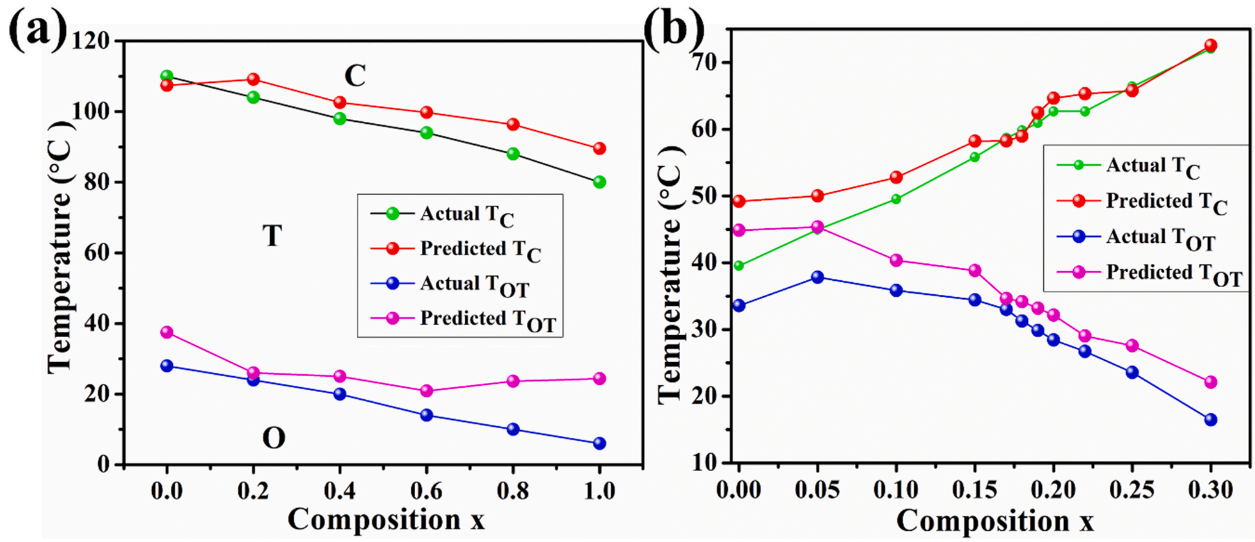

Fig. 4. Phase diagram construction from the other literatures (a) (1-x)$Ba_{0.95}Ca_{0.05}Ti_{0.95}Zr_{0.05}O_3$-(x)$Ba_{0.95}Ca_{0.05}Ti_{0.95}Sn_{0.05}O_3$ [48],and (b) (1-x)$BaTi_{0.89}Sn_{0.11}O_3$-x-$Ba_{0.7}Ca_{0.3}TiO_3$ [49].

model with assurance that the compositions are not present in the training dataset. The systems that we have considered for tentative composition-temperature phase diagram prediction in Fig. 4 are (1-x) $Ba_{0.95}Ca_{0.05}Ti_{0.95}Zr_{0.05}O_3$-(x)$Ba_{0.95}Ca_{0.05}Ti_{0.95}Sn_{0.05}O_3$ [48] (C1), and (1-x)$BaTi_{0.89}Sn_{0.11}O_3$-x$Ba_{0.7}Ca_{0.3}TiO_3$ [49] (C2) respectively. The composition C1 was used as a reference to predict the dielectric behaviour of the composition as a function of temperature which turns out to be well matching with the predicted behaviour in Fig. S4. The features considered in the present study successfully captured T_C phase transition temperatures in the spectrum (Fig. S4). In a similar way, the dielectric spectrum for C2 was predicted as well for different values of "x" in Fig. S5. To understand how well the model was able to predict the dielectric spectrum as a function of temperature for a given composition, the performance was evaluated using the cosine similarity between the actual spectrum and the predicted spectrum in Table S4. It is observed from Fig. S5 (C2) that not all compositions show a high value for cosine similarity between the actual and predicted curves. Even if, specifically for C2 compositions, the dielectric temperature spectrums have low cosine similarity, we were able to obtain a better estimate for the Curie temperatures for all compositions with the simple supervised model. As we have used the reference series (as in Refs. [48,49]) to validate the prediction capabilities of our model as the above compositions are not present in the dataset; the comparison between specific values of Curie transition temperatures is provided in the inset of Figure S4 and Figure S5. In the following section, Li-doped $BaTiO_3$ samples are synthesized, and their Curie temperatures and dielectric temperature spectrum are predicted.

4. Results and discussion

Now, to further experimentally validate our approach towards prediction, Li-doped $BaTiO_3$ ceramics were synthesized with varying weight percentages of Li (0 wt%, 2 wt%, and 4 wt%) using the solvothermal approach. The synthesis and characterization methods are detailed in the supplementary information file. The X-ray diffraction plots and lattice parameters for the synthesized samples are shown in Fig. 5 and Table 2.

During the Rietveld refinement, all the samples demonstrated a single-phase structure (tetragonal *p4mm*), and no impurity phase was detected. Further studies are required to conclude whether the samples consist of other phases to a small extent as well. As observed from Fig. S6, peak (110), slightly shifted towards lower angles upon increase of lithium percentage, which is consistent with the literature [50]. The ionic sizes of Li^+ ions are different for different coordination numbers, which is taken into account for the observed X-ray diffraction results [50]. With 12 coordination number, Ba^{2+} and Li^+ have ionic sizes of 0.161 nm and 0.118 nm, respectively, whereas with coordination number 6, Ti^{4+} and Li^+ have ionic sizes of 0.061 nm and 0.076 nm, respectively. Hence, the decrease in the lattice parameters upon adding Li^+ ions is due to the substitution of Ba^{2+} with Li^+, while the increase in them is because of the substitution of Li^+ at the B-site upon higher percentage doping.

Further, Fig. S7 shows the elemental analysis of the prepared $BaTiO_3$ sample which confirms the presence of barium and titanium in the composition. From the study, it is observed that the atomic percentages of barium and titanium are very close to each other, and no other impurities are present. Transmission electron microscopy results provide more insights into the structure of the synthesized $BaTiO_3$ nanoparticles, as shown in Fig. S8. Transmission electron microscopy results revealed the formation of nanoparticles of $BaTiO_3$ and 4 % Li–$BaTiO_3$ compositions with a diameter in the range of 30–100 nm. Inset in Fig. S8 revealed the SAED pattern for both samples, and the results display polycrystalline diffused diffraction rings, indicating the polycrystalline nature of the composition. Analysis of SAED patterns was carried out with the help of CrysTBox software [51].

In this section, the dielectric properties of the Li-doped $BaTiO_3$ samples were analyzed (Fig. S3). For dielectric measurements of the samples, the powders were pressed, and the pellets were sintered in a closed environment at 1100 °C for 2 h. During the sintering process, care was taken to minimize the loss of lithium from the compound due to its volatile nature by covering samples with crucibles. Investigating the imaginary component of electric modulus (M'') is critical for unravelling the material's electrical transport dynamics, and hopping mechanisms Figure S3 (a) [52]. The emergence of peaks in the M'' vs log(frequency) as well as tanδ vs log(frequency) plot reflects the relaxation processes occurring within the material. The dielectric permittivity exhibited pronounced frequency dependence at lower frequency region, gradually diminishing at higher frequencies. The high dispersion of relative permittivity in lower frequency limits is predominantly ascribed to the Maxwell-Wagner-Sillars (MWS) effect, also known as space-charge polarization [53,54]. At higher frequencies, the oscillation of the electric field is much faster as compared to the motion of the dipoles, hence the electric dipoles don't have enough time to align themselves along the electric field, resulting in the decrease of the dielectric constant

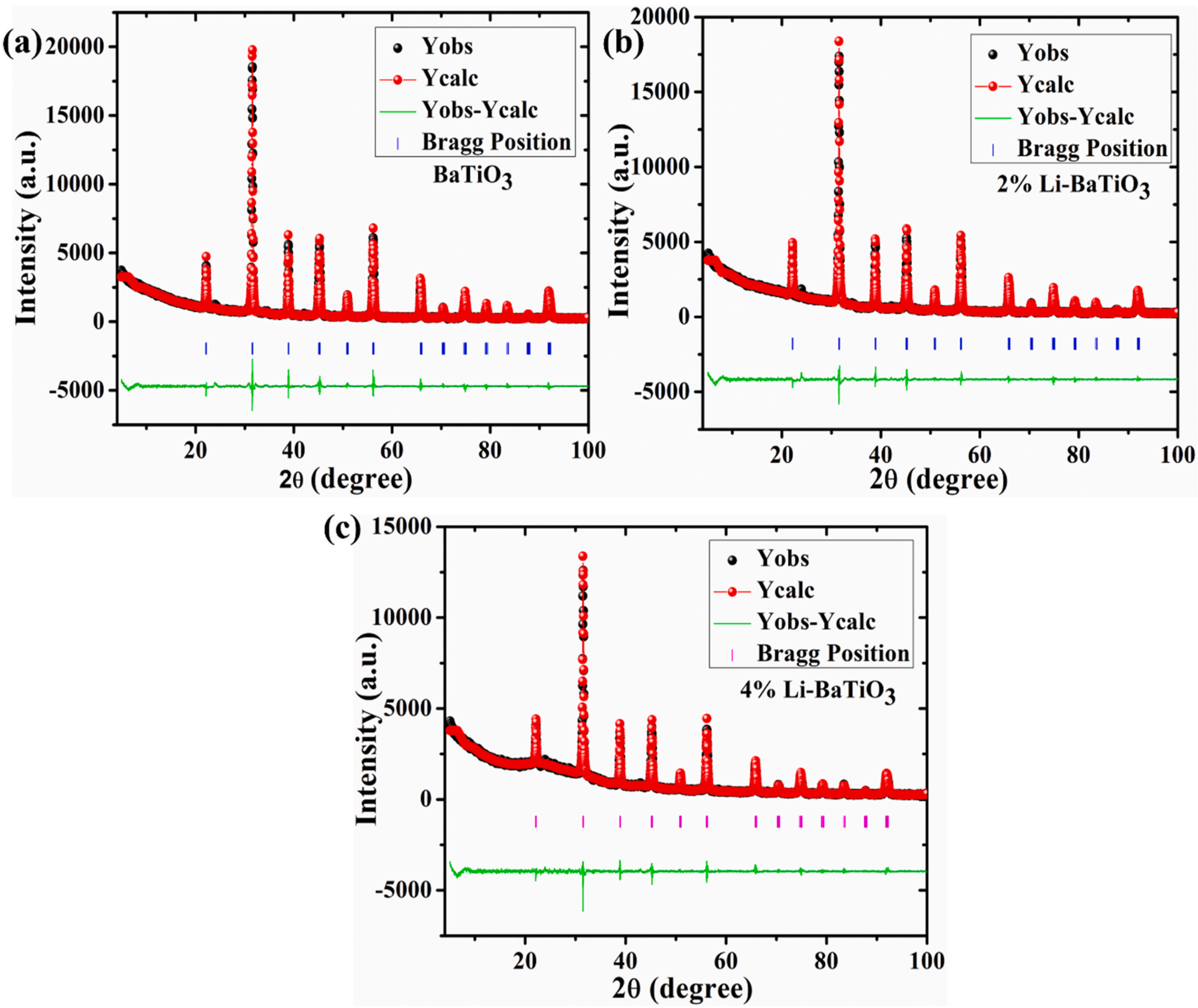

Fig. 5. Rietveld refinement of Li–$BaTiO_3$ compositions (a) $BaTiO_3$, (b) 2 % Li–$BaTiO_3$, and (c) 4 % Li–$BaTiO_3$.

Table 2
Lattice parameters for various Li-doped $BaTiO_3$ compositions.

Sample Name	a ($\AA$)	b ($\AA$)	c ($\AA$)	χ^2
$BaTiO_3$	4.0050	4.0050	4.0185	3.74
2 % Li–$BaTiO_3$	4.0061	4.0061	4.0154	3.15
4 % Li–$BaTiO_3$	4.0068	4.0068	4.0151	2.53

Figure S3 (b). In the overall range of frequency, 4 % Li–$BaTiO_3$ showed relatively less dielectric loss as compared to other compositions Figure S3 (c).

The temperature-dependent dielectric spectrum of all samples was captured with the help of a dielectric impedance spectroscope at 10 MHz, and the prediction of the dielectric plot was carried out. The dielectric results of the Li–$BaTiO_3$ samples are consistent with the earlier literature [50]. As observed, the predictions are in good agreement with the actual results except for the 2 % Li–BaTiO3 sample (Fig. 6). The Curie temperature for all the synthesized samples is tabulated in Table 3, along with predicted values from the ML algorithms. A reason for the formation of polar nano-region is provided for a slight increase in the Curie temperature upon an increase in the lithium concentration [50].

As in the given study, we predicted T_C and T_{OT} for lead-free compositions and also extended the same ML model for Li-doped $BaTiO_3$ systems. As mentioned, it is clearly visible from the feature importance, T_C is highly influenced by the nature of elements at the B-site of the crystal structure. In the case of size factors, the atomic volume of B-site elements and CRSB are important and directly influence the piezoelectric properties [22,35]. Among electrochemical properties, ENB is critical as ferroelectric properties gets highly influenced by the electronegativity of the elements.

In the present section, we employed machine learning-based reverse engineering to optimize the synthesis process of lead-free ceramics, utilizing a combination of features denoted as VA and VB. The color scheme in Fig. 7, with red and blue representing high and low values of the target property, visually depicts the distribution of the target property. The comparison revealed the significant overlap between the predicted and the actual high-value regions in the contour plot for all the considered target properties [44]. The above observation strongly suggests that the employed machine learning regression model successfully captured the underlying relationship between the input features (VA and VB), and the target properties during the training phase, and the training allowed the model to replicate it well during the prediction of the unseen data. The given strategy paves the way for extending the approach to

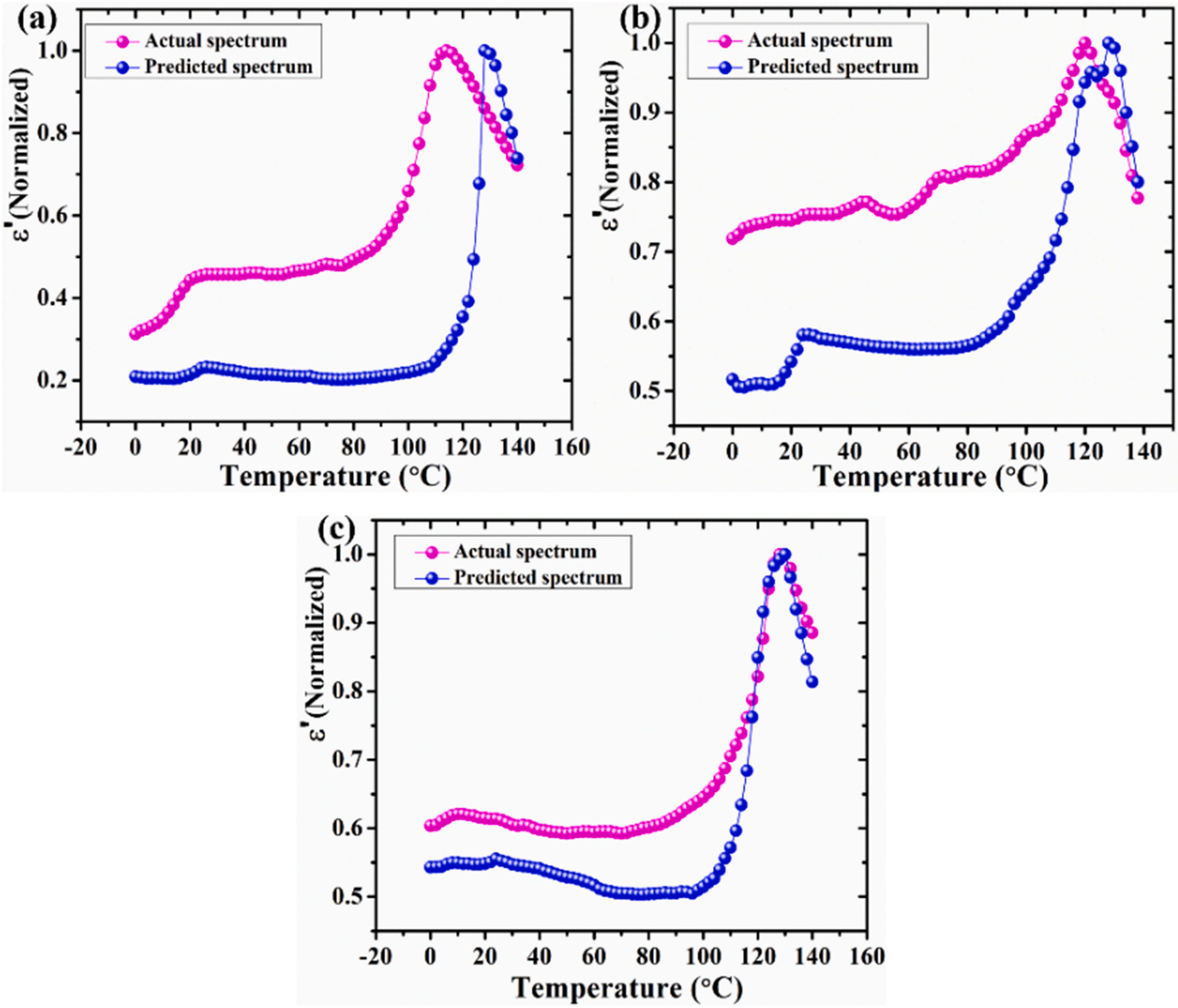

Fig. 6. Prediction vs actual spectrum of different Li–$BaTiO_3$ compositions (a) 0 % Li–$BaTiO_3$, (b) 2 % Li–$BaTiO_3$, and (c) 4 % Li–$BaTiO_3$.

Table 3
Prediction of Curie temperature for synthesized Li-doped BT samples.

Sample Name	Actual T_C (°C)	Predicted T_C (°C) [XG-boost]	Predicted T_C (°C) [XG-boost] [from Fig. 6]	Max % error
$BaTiO_3$	114	122.2	127	11.40
2 % Li–$BaTiO_3$	120	116.6	127.5	6.25
4 % Li–$BaTiO_3$	128	116.8	128.7	8.75

predict the optimal range for other features created within the given model. Consequently, the insights gleaned from the generated feature-based contour plots possess broader applicability, potentially extending to various piezoelectric ceramic systems [44]. In conclusion, the XG-boost model not only delivers optimal performance but also unveils new physical insights through its effective feature engineering capabilities. A similar analysis based on the contour plots with features CRSA and CRSB for given properties is provided in Fig. S9.

With the help of the developed regression model, we attempted to understand the variation of T_C and T_{OT} over a wide range of compositions given as $(Ba_{1-x-y}Ca_xLi_y)$ $(Ti_{1-u-v-w}Zr_uSn_vHf_w)$ O_3. The values of all the variables vary in the range as, $0 < x, y < 0.3$, and $0 < u, v, w < 0.2$ with a step size of 0.05. As the stoichiometric coefficients vary, new compositions will be created, and the required features for each composition will be stored. A python program is developed, giving a total compositional space of around 6150. The constraints for stoichiometry are provided as, $x + y \leq 0.3$ and $u + v + w \leq 0.75$, respectively to avoid the other phase formation. The trained model with Dataset 1 predicted the functional properties for the complete compositional space created, and the 3D scatter plots were plotted and shown in Fig. 8. The new feature maps cover large compositional space and provide a broader range of feature values to understand the effect of various dopants on the target property. All 1400 computationally created compositions were tabulated and sorted to trace back the element-wise stoichiometry of the composition, allowing us to optimize the synthesis process with a specific range for target property. An approach was implemented by J. He et al. to find the optimized composition of piezoelectric ceramic with the highest value of d_{33} [4]. In the present approach, it is possible to identify a few compositions with the desired range for the transition temperature. For example, if a specific application needs ceramics with T_C in the range of 90–100 °C, the present machine learning model can suggest several compositions

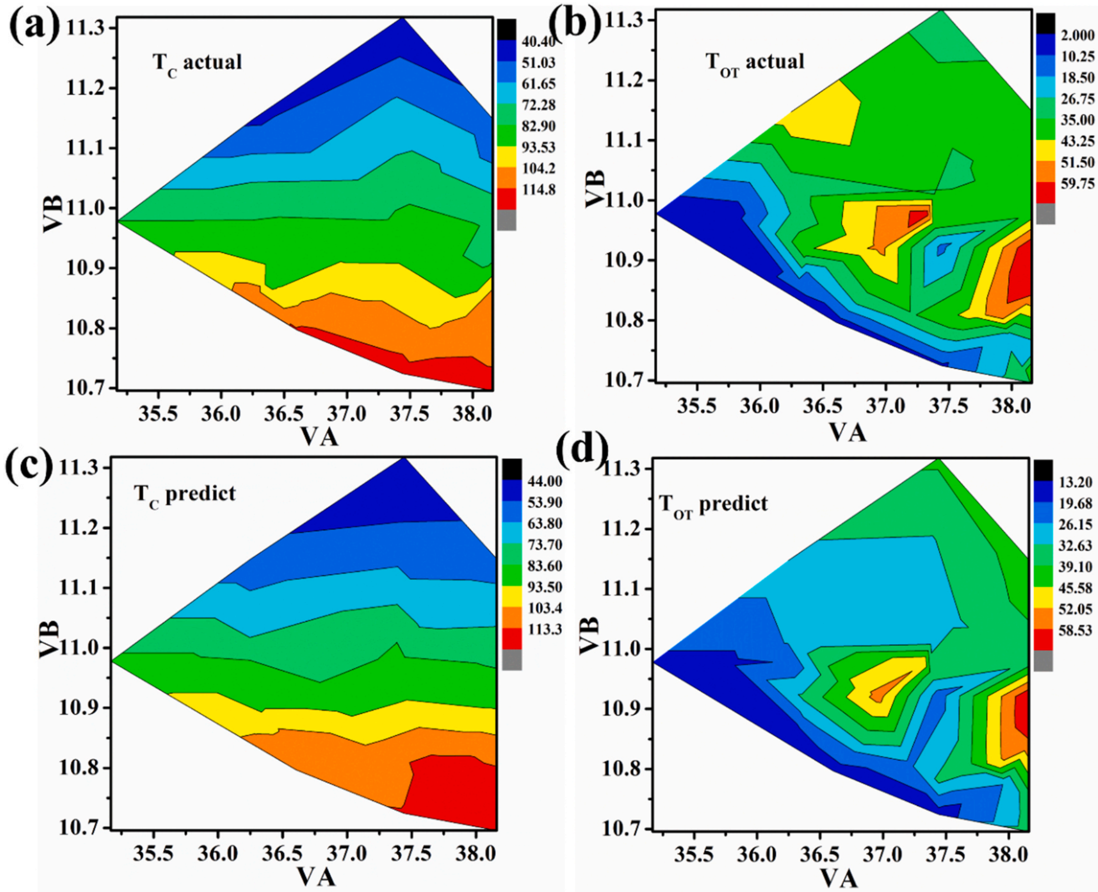

Fig. 7. Contour plots for target properties in the VA and VB space for reverse engineering (a, c) actual T_C contour plot and predicted T_C contour plot, (b, d) actual T_{OT} contour plot and predicted T_{OT} contour plot (unit: °C).

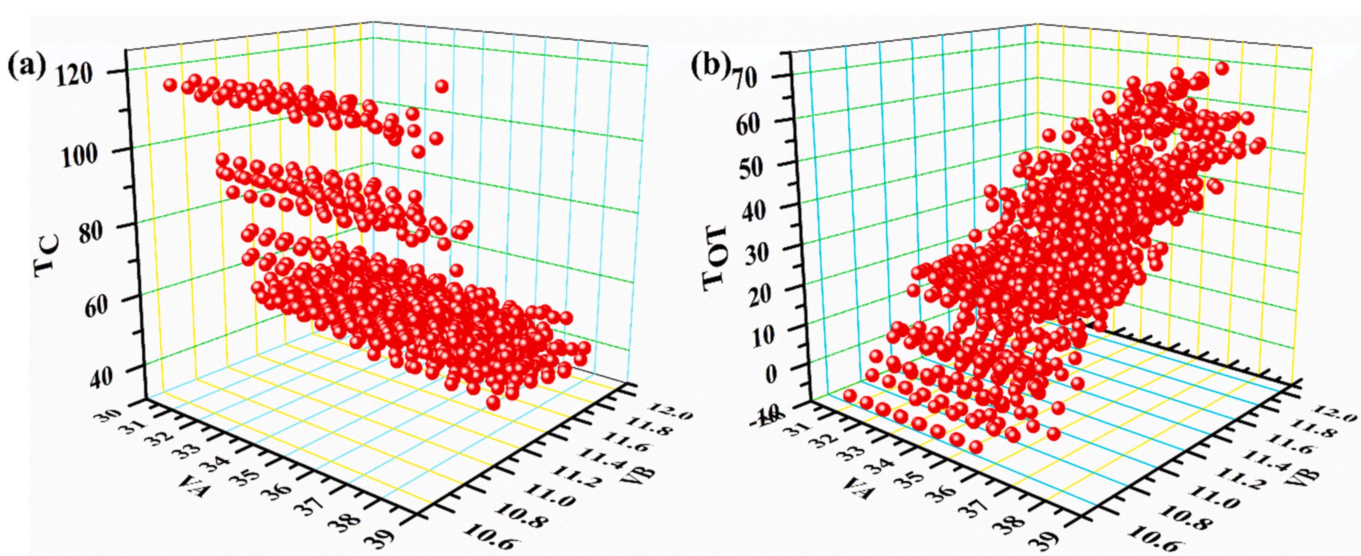

Fig. 8. 3D scatter plots as a function of features VA and VB for generated compositions to understand their behaviour with respect to the features and doping of various elements at A-site and B-site (a) T_C, (b) T_{OT}.

having Curie temperature in the given range. The same approach can be extended to other properties such as, d_{33}, energy storage properties, and ferroelectric properties for the given system. As analyzed, all the scatter plots demonstrate the strong influence of B-site elemental doping on the ferroelectric properties of lead-free ceramics and are consistent with the literature. Finally, for reference purposes, all the element-wise properties, along with their units, are provided in Table S5, along with the python program in the supplementary information.

5. Conclusions

Machine learning-motivated framework was proposed to forecast the phase-transition temperatures and dielectric temperature spectrum for the $(Ba_{1-x-y}Ca_xLi_y)$ $(Ti_{1-u-v-w}Zr_uSn_vHf_w)$ O_3 system. Among the three machine learning models (XG-boost, random forest, and SVR-rbf) used, XG-boost demonstrated better prediction capabilities with standard deviation for T_C as $\Delta T_C = 5.87\ ^\circ C$, and for T_{OT} as $\Delta T_{OT} = 7.53\ ^\circ C$. Also, a total of 17 dielectric temperature spectrums were predicted for different complex compositions, with a mean absolute error of 8.1 °C in the T_C values. Again, a maximum error of 11.4 % was obtained in the predicted T_C value of the synthesized A-sight Li-doped $BaTiO_3$ samples. Finally, the concept of reverse engineering is extended to optimize the synthesis of lead-free ceramics by tuning the specific range for features as well as elemental stoichiometry. With the help of reverse engineering, we can pinpoint the required stoichiometry of lead-free ceramics to achieve the desired value of transition temperature. The reported study provides new insights for feature engineering, which successfully captured the T_C and T_{OT} phase transition temperatures in the predicted dielectric spectrum.

Data statement

Further programs and data will be available from the corresponding author upon request.

CRediT authorship contribution statement

Srujan Sapkal: Methodology, Investigation, Formal analysis, Conceptualization. **Balasubramanian Kandasubramanian:** Writing – review & editing, Supervision, Resources, Conceptualization. **Himanshu Sekhar Panda:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.matchemphys.2024.129999.

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